SENSORS INSIDE OUT

Sensing, Computing and Actuation Hardware for a Driverless Formula Student Car

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ABSTRACT

A driverless race car has no human driver to look at the track, judge the speed or press the pedals. All of this has to be done by sensors, a computer and actuators. This paper provides a description of the sensors and hardware used on a Formula Student Driverless car. It explains the Hesai Pandar40P LiDAR, the VectorNav VN-200 GNSS/INS, the Stereolabs ZED 2i stereo camera (and how it can be used as a mono camera), the NVIDIA Jetson AGX Orin computer, the IMU and INS, wheel speed sensors, steer-by-wire and brake-by-wire systems, the different types of accelerometers and magnetometers, Hall effect sensors and optical sensors. For each sensor the working principle, the common errors and how to solve them are discussed. Finally the paper explains how all these parts are integrated into the car and where they can be sourced from in the Indian market. The objective is to understand how each sensor fits into the autonomous system and why it is needed.

Keywords— LiDAR, GNSS/INS, IMU, Stereo Camera, Jetson AGX Orin, Sensor Fusion, Hall Effect, Wheel Speed Sensor, Steer-by-Wire, Brake-by-Wire, Formula Student Driverless

I. INTRODUCTION

In self-driving car the car has to drive around a track that is marked only by traffic cones. In the standard rules the blue cones mark the left edge, the yellow cones mark the right edge and orange cones mark the start and finish area. The car does not know the track before the run, so it has to see the cones, find out where it is, plan a path and then steer, accelerate and brake on its own. The dynamic events are acceleration, skidpad, autocross and trackdrive.

The software that does this is usually split into four parts: perception (finding the cones), localization and mapping or SLAM (finding where the car is and building a map of cones), planning (finding the best path) and control (turning that path into steering, throttle and brake commands). Every one of these parts depends on good sensor data. If the sensors are wrong, even the best algorithm will make the car go off track. This is why choosing the right sensors, removing their errors and mounting them properly is one of the most important jobs in a driverless team.

II. SENSOR SUITE OVERVIEW

Table I gives a short summary of all the hardware that is discussed in this paper and what each part is used for. The sensors can be divided into exteroceptive sensors, which look at the world outside the car (LiDAR, camera), and proprioceptive sensors, which measure the car's own motion (IMU, wheel speed, steering angle).

TABLE I
SENSOR SUITE OF THE DRIVERLESS CAR

Hardware	Measures	Main job
Hesai Pandar40P	3D point cloud	Cone detection, range
ZED 2i	Colour image, depth	Cone colour, depth
VectorNav VN-200	Position, velocity, attitude	State estimation
IMU (in VN-200 / ZED)	Acceleration, angular rate	Motion, attitude
Wheel speed sensors	Wheel RPM	Speed, slip, odometry
Steering encoder	Steering angle	Steering feedback
Jetson AGX Orin	—	Runs all software
Steer/brake-by-wire	—	Actuation, EBS

III. LIDAR – HESAI PANDAR40P

A. How LiDAR works

LiDAR stands for Light Detection and Ranging. The sensor sends a short laser pulse and measures the time t it takes for the pulse to hit an object and come back. Since light travels at speed c, the distance d is:

$$d = c \cdot t / 2 \qquad (1)$$

The factor 2 is because the light travels to the object and back. A spinning LiDAR has many lasers stacked vertically and rotates 360°, so in one rotation it gives a full 3D picture of the surroundings called a point cloud. Each point has x, y, z coordinates and an intensity value, which tells how reflective the surface was.

B. The Pandar40P

The Pandar40P is a 40-channel mechanical spinning LiDAR from Hesai. Its main specifications from the datasheet are given in Table II. An important point is that its laser channels are not spread evenly: the channels are packed closely near the horizon and are spread out towards the top and bottom. This is useful for self-driving cars because the cones are small and are always near the ground plane ahead of the car.

C. Use in the car

The point cloud is first cropped to a region in front of the car. Then the ground is removed using a plane fitting method such as RANSAC, and the remaining points are grouped into clusters (for example with Euclidean clustering or DBSCAN). Clusters that have the size and shape of a cone are kept as cone candidates. The LiDAR gives very accurate distance to the cone, but it cannot reliably tell the colour of the cone, so its output is fused with the camera.

D. Errors and how to solve them

Motion distortion: one rotation takes 100 ms at 10 Hz. At 20 m/s the car moves 2 m during one scan, so the cloud gets smeared. This is solved by *deskewing*, where each point

is corrected using its own timestamp and the IMU/INS motion. **Low point density at range:** a cone 15 m away may only be hit by 2–3 rings, so cone detection range is limited; mounting the LiDAR low and tilting it slightly helps. **Calibration errors:** a small rotation error in the LiDAR mount causes large position errors far away, so the LiDAR-to-car (extrinsic) transform must be calibrated. **Time sync:** the LiDAR timestamps are synchronised to GNSS time using the PPS signal or PTP. **Rain, dust and reflections:** these create noise points, which are removed with intensity and outlier filters, and the dual-return mode can be used.

IV. IMU AND INS

A. Inertial Measurement Unit (IMU)

An IMU contains a 3-axis accelerometer and a 3-axis gyroscope, and often a 3-axis magnetometer. The accelerometer measures specific force (acceleration plus gravity) and the gyroscope measures angular rate. The IMU only gives raw measurements; it does not directly tell where the car is.

B. Inertial Navigation System (INS)

An INS takes the IMU measurements and integrates them: angular rate is integrated once to get orientation, and acceleration is integrated twice to get velocity and then position. The problem is that any small error also gets integrated. A constant accelerometer bias b gives a position error that grows with the square of time:

$$e(t) = \tfrac{1}{2} \cdot b \cdot t^2 \qquad (2)$$

For example, a bias of only 0.01 m/s² gives an error of $0.5 \times 0.01 \times 60^2 = 18$ m after one minute. This is why a pure INS drifts and why it is always aided by GNSS, wheel speed or LiDAR. A GNSS-aided INS like the VN-200 runs a Kalman filter to combine the fast but drifting IMU data with the slow but absolute GNSS data.

TABLE III
IMU VS INS

	IMU	**INS**
Output	Raw acceleration and angular rate	Position, velocity, attitude
Processing	None or basic filtering	Integration + Kalman filter
Drift	Bias and noise in raw data	Grows with time unless aided
Example	IMU inside ZED 2i	VectorNav VN-200

C. IMU errors and how to solve them

Bias: an offset present even when the car is still. It is estimated at start-up by keeping the car stationary for a few seconds, and online by the Kalman filter. **Bias instability and random walk:** the bias slowly wanders; these values are found using an *Allan variance* test (record the IMU still for a few hours and plot the Allan deviation) and are used to tune the filter noise. **Scale factor and misalignment:** fixed by calibration on a level surface or turntable. **Temperature:** the bias changes with temperature, so the IMU should be kept away from the motor, inverter and Jetson exhaust. **Vibration:** engine/motor and road vibration cause aliasing and noise, so the IMU should be mounted on a stiff bracket with proper damping and a suitable low-pass filter. **Lever arm:** if the IMU is not at the reference point of the car, the offset must be measured and entered, otherwise rotation is seen as false acceleration.

V. VECTORNAV VN-200 GNSS/INS

The VN-200 is a small GNSS-aided inertial navigation system. Inside it there is a MEMS IMU (3-axis accelerometer, gyroscope and magnetometer), a barometric pressure sensor and a GNSS receiver. It runs an Extended Kalman Filter (EKF) on board, so it directly outputs position, velocity and attitude (roll, pitch, yaw) at a high rate. The rugged version comes in an aluminium enclosure with a connector, which is better for a race car.

A. Errors and how to solve them

Heading at standstill: the VN-200 has a single GNSS antenna, so it gets heading from the magnetometer when still and from GNSS velocity when moving. In an electric car the magnetometer is disturbed by the motor, battery and HV cables, so heading can be wrong until the car starts moving. Solutions are to do a hard/soft iron calibration in the car, use the VN-200's dynamic heading mode, or move to a dual-antenna unit (VN-300) if heading at rest is needed. **GNSS position is only metre-level:** this is not enough to drive between cones, so in FSD the INS is mainly used for velocity and attitude, while the cone map from SLAM gives the position. An RTK correction can be added for testing. **Multipath and blocked sky:** the antenna should be on top of the car with a ground plane and a clear view of the sky. **Lever arm:** the antenna offset from the IMU must be entered in the configuration.

VI. TYPES OF ACCELEROMETERS

An accelerometer is basically a small mass (proof mass) on a spring. When the sensor accelerates, the mass lags behind and the spring stretches by an amount proportional to the acceleration. The different types differ in how they measure this movement.

A. Capacitive MEMS

The proof mass forms one plate of a capacitor with comb-like fingers. Movement changes the gap and so the capacitance. These are tiny, cheap, low power and can measure down to 0 Hz (static gravity), which is why they are used in almost every IMU, including the VN-200 and ZED 2i.

B. Piezoelectric

A piezoelectric crystal produces a charge when it is squeezed by the mass. They have very high bandwidth and are great for vibration and shock testing, but they cannot measure static or very slow acceleration, so they are not used for navigation.

C. Piezoresistive

Strain gauges on a beam change resistance when the beam bends. They can measure static acceleration and high shock, and are used for crash testing and high-g measurements.

D. Servo (force-balance)

A feedback coil keeps the mass in the centre, and the current needed is proportional to the acceleration. These are

very accurate and stable and are used in high-end navigation and aircraft, but they are big and costly.

E. Thermal (convective)

A heated gas bubble shifts inside a sealed cavity when accelerated, and temperature sensors detect the shift. There are no moving parts, so they are shock resistant, but bandwidth is low.

TABLE V
COMPARISON OF ACCELEROMETER TYPES

Type	Static (0 Hz)	Bandwidth	Typical use
Capacitive MEMS	Yes	Low–medium	IMUs, phones, INS
Piezoelectric	No	Very high	Vibration, NVH
Piezoresistive	Yes	High	Crash, shock
Servo	Yes	Low	Precision navigation
Thermal	Yes	Low	Tilt, rugged use

VII. TYPES OF MAGNETOMETERS

A magnetometer measures the magnetic field. In a car it is used as an electronic compass to find heading from the Earth's field, which is very weak compared to the fields near motors and current-carrying cables.

A. Hall effect

Uses the Hall voltage (Section VIII). Simple and cheap but has low sensitivity, so it is better for strong fields like position and current sensing than for compassing.

B. Magnetoresistive (AMR, GMR, TMR)

The resistance of a thin film changes with the magnetic field. Anisotropic (AMR), Giant (GMR) and Tunnel (TMR) magnetoresistance give increasing sensitivity. AMR and TMR chips are common in IMUs because they are small and sensitive enough for the Earth's field.

C. Fluxgate

A soft magnetic core is driven in and out of saturation by a coil, and a pick-up coil measures how the external field shifts this. Very accurate and low noise, used in survey and space, but large and power hungry.

D. Magneto-inductive

The inductance of a coil changes with the external field and is measured as a change in oscillator frequency. Good resolution at low power.

E. Magnetometer errors

Hard-iron error is a constant offset caused by permanent magnets and steel near the sensor. **Soft-iron** error distorts the field into an ellipse and is caused by nearby ferrous material. Both are calibrated by rotating the sensor (or the whole car) through all directions and fitting an ellipsoid, which gives an offset vector and a correction matrix. In an EV, the motor current changes the field while driving, which calibration cannot remove, so the magnetometer should be mounted far from HV parts and given low trust in the filter.

VIII. HALL EFFECT SENSORS

When a current I flows through a thin conductor or semiconductor of thickness t and a magnetic field B is applied perpendicular to it, the charge carriers are pushed to one side by the Lorentz force. This creates a small voltage across the plate called the Hall voltage:

$$V_H = I \cdot B / (n \cdot q \cdot t) \qquad (3)$$

where n is the carrier density and q is the charge of a carrier. Since V_H is proportional to B, the sensor can detect a magnet or a steel tooth passing by. Hall sensor ICs come in four main types: **unipolar** switches (turn on near one pole), **bipolar** switches, **latching** switches (turn on with north, off with south) and **linear** sensors (analog output proportional to field). Hall sensors are contactless, work down to zero speed and are not affected by dirt, which makes them ideal for a race car. On our car they are used in wheel speed sensors, in the BLDC motor of the steering actuator for commutation, in pedal and throttle position sensors and in current sensors for the low-voltage system.

Errors: the output drifts with temperature and depends strongly on the air gap to the magnet or tooth. The solution is to use sensors with built-in temperature compensation, fix the air gap with a rigid bracket and shims, and use shielded twisted-pair cables to avoid EMI from the inverter.

IX. WHEEL SPEED SENSORS

A wheel speed sensor counts teeth on a toothed ring (tone wheel or trigger wheel) that rotates with the wheel. If the ring has N teeth and the sensor measures a pulse frequency f, the wheel speed and car speed are:

$$\omega = 2\pi f / N, \quad v = \omega \cdot R \qquad (4)$$

where R is the effective rolling radius of the tyre. There are two main types: **passive** variable reluctance (VR) sensors, which use a coil and magnet and produce an AC signal whose amplitude drops at low speed, and **active** sensors (Hall effect or magnetoresistive) which give a clean square wave down to zero speed. For a driverless car active sensors are preferred, because the car must know its speed accurately when starting and stopping.

Wheel speed is used for odometry in the state estimator, for slip and traction control, and as a safety check (for example, confirming the car has actually stopped).

A. Errors and how to solve them

Wheel slip: under hard acceleration or braking the wheel speed is not the car speed. The fix is to use the non-driven wheels or the slowest wheel during acceleration and to give wheel odometry low trust in the EKF when slip is detected. **Low resolution at low speed:** with few teeth, pulses are far apart; measure the period between pulses instead of counting pulses in a fixed window, or use more teeth. **Tyre radius changes:** radius changes with pressure, temperature and load, so it should be calibrated by driving a known distance. **Air gap and EMI:** use a rigid mount, check the gap with a feeler gauge, and use shielded cables and input filtering at the microcontroller.

X. OPTICAL SENSORS

Optical sensors use light to measure position, speed or temperature without contact. The ones that are useful in a Formula Student car are given below.

A. Optical encoders

A disc with slots or a printed pattern rotates between an LED and a photodetector. **Incremental** encoders give pulses (A/B channels in quadrature, so the direction is also known) and need a homing step after power-on. **Absolute** encoders give a unique code for every angle, so the angle is known immediately at power-on. For the steering angle of a driverless car, an absolute encoder is preferred because the car must know where the wheels are pointing as soon as it is switched on.

B. Optical ground speed sensors

These sensors look down at the road and measure the real speed over ground using the movement of the road texture (correlation optics). They measure both longitudinal and lateral speed, so they give the true slip angle and are very useful for validating the state estimator. They are expensive, so teams usually borrow or rent them for testing.

C. Infrared temperature sensors

Infrared thermopile sensors measure tyre and brake disc temperature without contact, which helps in tuning the car.

D. Errors

Optical sensors are affected by dirt, mud and water on the lens or disc, and by direct sunlight. They should be sealed (IP rated), cleaned before every run and shielded from the sun.

XI. STEREO ZED 2I CAMERA (AND MONO USE)

A. Stereo vision

The ZED 2i from Stereolabs has two cameras side by side, separated by a baseline B of 12 cm. The same cone appears at slightly different positions in the left and right images, and this shift is called the disparity d. With focal length f (in pixels) the depth is:

$$Z = f \cdot B / d \qquad (5)$$

Because the disparity becomes very small for far objects, the depth error grows with the square of the distance:

$$\Delta Z \approx (Z^2 / (f \cdot B)) \cdot \Delta d \qquad (6)$$

So the ZED depth is good up close but becomes less reliable at long distances, where the LiDAR is much better. The ZED 2i also has a built-in IMU, barometer, magnetometer and temperature sensors, and the SDK can run visual-inertial odometry.

B. Using it as a mono camera

A mono camera is a single camera. It gives a colour image but no direct depth. In many FSD pipelines the camera is only needed for one thing that the LiDAR cannot do: telling the colour of the cone. In that case we can use only the left image of the ZED 2i as a mono camera. A neural network such as YOLO runs on the Jetson and draws boxes around blue, yellow and orange cones. Depth for a mono detection can be found in two ways: from the known height of the cone ($Z \approx f \cdot H_{\text{cone}} / h_{\text{pixels}}$) or, better, by projecting the LiDAR cones into the image and matching them with the boxes (camera-LiDAR fusion). Using mono mode also saves GPU time, since stereo depth computation is not needed.

C. Errors and how to solve them

Lens distortion and intrinsic errors: calibrate with a checkerboard (Zhang's method), although the ZED comes factory calibrated. **Camera-LiDAR extrinsic error:** calibrate using a target seen by both sensors and check by overlaying the point cloud on the image. **Sunlight, glare and shadows:** use the polariser lens option, auto exposure and train the network with images in different light. **Motion blur:** use a short exposure. **Vibration:** use a rigid mount, since even small movement of the camera changes the calibration. **Latency:** timestamp every frame and compensate for delay in the fusion.

XII. NVIDIA JETSON AGX ORIN

The Jetson AGX Orin is the main computer of the car. It is an embedded module with an Arm CPU and an NVIDIA Ampere GPU with Tensor cores, so it can run neural networks and point cloud processing in real time while using much less power than a desktop PC. It runs Linux (JetPack) and supports CUDA, TensorRT and ROS 2.

TABLE VII
JETSON AGX ORIN MODULES (PER NVIDIA DATASHEET)

Parameter	AGX Orin 64GB	AGX Orin 32GB
AI performance	Up to 275 TOPS	Up to 200 TOPS
CPU	12-core Cortex-A78AE	8-core Cortex-A78AE
GPU	2048 CUDA, 64 Tensor	1792 CUDA, 56 Tensor
Memory	64 GB LPDDR5	32 GB LPDDR5
Power	15–60 W	15–40 W

On the car, the Orin runs the LiDAR and camera drivers, cone detection (YOLO with TensorRT), camera-LiDAR fusion, SLAM, the EKF state estimator, path planning and the controller (for example Pure Pursuit or MPC). It sends steering, throttle and brake requests over CAN.

A. Errors and how to solve them

Thermal throttling: in a hot sidepod the Orin reduces its clock speed. It needs good airflow and its power mode should be set with `nvpmodel`, with `jetson_clocks` used to fix clocks. **Power brownouts:** voltage dips when the actuators draw current can reset the Orin, so it should be powered through its own DC-DC converter with input filtering and hold-up capacitors. **Storage corruption:** sudden power cuts can corrupt the disk, so use an NVMe SSD for logs and a controlled shutdown. **Software timing:** use real-time priorities, separate CPU cores and monitor the loop time; a watchdog on the controller side must stop the car if the Orin stops sending messages.

XIII. STEER-BY-WIRE AND BRAKE-BY-WIRE

In a normal car the driver's hands and feet are mechanically connected to the steering and brakes. In a driverless car these are replaced by electric actuators that are controlled by commands, which is called *by-wire*. In Formula Student the car must still be drivable by a human in manual mode, so the actuators are usually added on top of the normal mechanical system.

A. Steer-by-wire

A motor (typically a BLDC or servo motor with a gearbox, belt or gear drive) is connected to the steering column or rack. The motor controller runs a position loop

using feedback from an absolute steering encoder. The Orin sends the desired steering angle over CAN. It must be possible to disengage the actuator (for example with a clutch) for manual driving.

B. Brake-by-wire and EBS

The service brake actuator presses the brake pedal or pushes on the master cylinder, using an electric linear actuator or a pneumatic cylinder, with brake pressure sensors for feedback. In addition, the rules require an Emergency Brake System (EBS) that brings the car to a stop automatically if anything fails, for example if power is lost or the Remote Emergency System (RES) is pressed. The EBS is normally built to be *fail-safe*: stored energy (compressed air or a spring) is held back by a valve that needs power to stay closed, so losing power applies the brakes.

C. Errors and safety

Actuator lag and backlash: identify the actuator response and include it in the controller, and reduce backlash in the mechanical linkage. **Saturation:** limit the commanded rate and angle. **Communication loss:** use a heartbeat on CAN with timeouts; if no valid command arrives in time, the system goes to a safe state. **Sensor failure:** check plausibility between redundant sensors (for example two brake pressure sensors). **Startup checks:** the EBS must be tested automatically before the car is allowed to go to the ready state, as described in the competition rules.

XIV. SENSOR FUSION AND TIME SYNCHRONISATION

No single sensor is good enough alone. The LiDAR is accurate but slow (10–20 Hz), the IMU is fast but drifts, the wheel speed is good but suffers from slip, and GNSS is absolute but noisy. An Extended Kalman Filter combines all of them: it *predicts* the car's state using the IMU and a vehicle model, and *corrects* the prediction whenever a new measurement arrives, weighting each sensor by how much it is trusted (its covariance).

For fusion to work, all sensors must share the same time and the same coordinate frames. Time is synchronised by sending the GNSS PPS pulse and NMEA time to the LiDAR, and using PTP or NTP on the Ethernet network so the Orin and LiDAR agree. Frames are handled in ROS 2 with the `tf2` tree, which stores the measured position and rotation of every sensor relative to the car's base frame. If timestamps are off by even 50 ms, a cone seen at 20 m/s will be placed 1 m away from where it really is.

XV. SUMMARY OF ERRORS AND SOLUTIONS

TABLE VIII
COMMON SENSOR ERRORS AND SOLUTIONS

Sensor	Error	Solution
LiDAR	Motion distortion	Deskew with IMU/INS
LiDAR	Mount misalignment	Extrinsic calibration
Camera	Distortion, glare	Checkerboard calib., polariser
Camera	Depth error at range	Fuse with LiDAR
IMU	Bias, drift	Static init, Allan variance, EKF
IMU	Vibration	Rigid mount, damping, filter
Magnetometer	Hard/soft iron, EV fields	Ellipsoid calib., distance
GNSS/INS	Multipath, no heading	Clear sky, dynamic heading
Wheel speed	Slip, radius change	Slip detect, radius calib.
Hall sensor	Air gap, temperature	Rigid mount, compensated IC
Optical	Dirt, sunlight	Sealing, cleaning, shading
Jetson	Throttling, brownout	Cooling, own DC-DC
By-wire	Comm. loss	Heartbeat, fail-safe EBS
All	Time offset	PPS / PTP sync, timestamps

XVI. INTEGRATION IN THE CAR

Fig. 1 shows how all the hardware is connected. The Orin is the centre of the high-level system and the autonomous system controller (or VCU) is the bridge to the actuators and safety system.

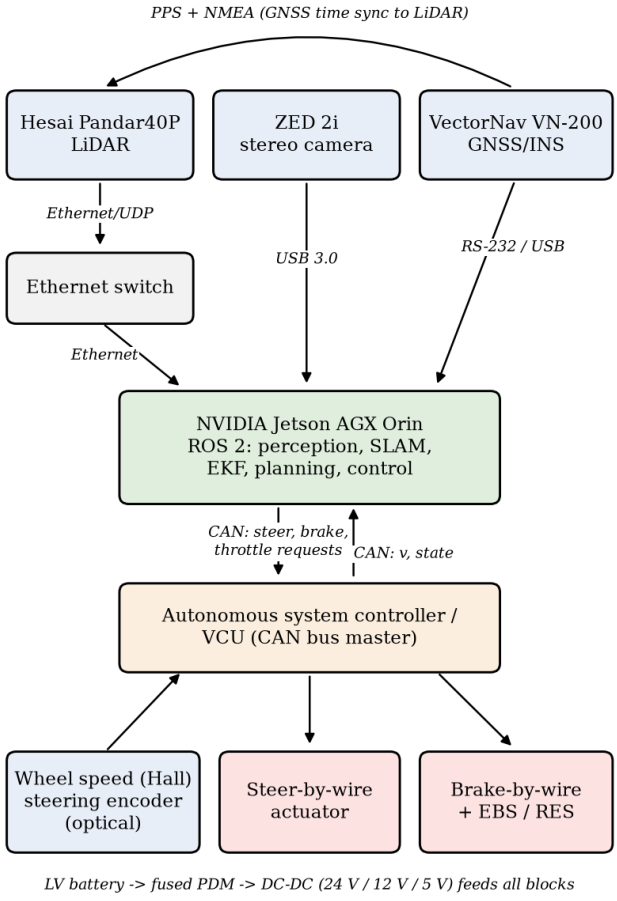


Fig. 1. Hardware architecture of the driverless car showing the data links between the sensors, the Jetson AGX Orin, the CAN-based controller and the actuators.

A. Mounting positions

LiDAR: usually on the front of the car (on the front wing or nose) at a low height, so that more laser rings hit the small cones. A higher mount on the main hoop gives a 360° view but loses near cones. **Camera:** high on the main hoop or above the driver, looking forward, so it can see over the nose. **VN-200:** close to the centre of gravity on a stiff part of the chassis, with its axes aligned to the car's axes. **GNSS antenna:** on top of the car with a clear sky view. **Wheel speed sensors:** on the uprights, facing a tone wheel on the hub. **Steering encoder:** on the steering column or rack. All

mounts should be stiff, because if a bracket bends the calibration becomes wrong.

B. Power

The low-voltage (LV) battery feeds a power distribution module with a fuse for each branch. Separate DC-DC converters supply the Orin, the sensors and the actuators so that actuator current spikes do not reset the computer. The total power of the autonomous system (Orin up to 60 W, LiDAR ≈18 W, camera, INS, actuators) must be checked against the LV battery capacity for the length of the trackdrive event.

C. Communication

The LiDAR connects through a small automotive Ethernet switch, the ZED 2i through a locking USB 3.0 cable, and the VN-200 through serial (or a serial-to-USB adapter). The Orin's CAN port connects to the car's CAN bus, which is terminated with 120 Ω resistors at both ends. Cables should be shielded, kept away from HV cables, and use locking automotive connectors that can handle vibration.

D. Software

Every sensor has a ROS 2 driver that publishes its data with timestamps. The `tf2` tree stores the sensor positions. All data is recorded using `ros2 bag` during testing, so that the same run can be replayed later to tune the algorithms without the car.

E. Protection and testing

Sensors must be protected from water, dust and heat (IP rated parts, covers and cooling). Testing should be done step by step: first each sensor on the bench, then on the car while stationary, then at low speed, and only then at full speed.

XVII. SOURCING IN THE INDIAN MARKET

Most of the high-end sensors are not made in India, so they are either imported directly or bought through Indian distributors and resellers. Table IX lists possible sources. Prices and availability change often, so they must be confirmed with the vendor before ordering.

TABLE IX
POSSIBLE SOURCES IN INDIA

Item	Possible sources
Hesai Pandar40P	Hesai sales team / authorised distributors
VectorNav VN-200	VectorNav direct or its authorised distributors (import)
ZED 2i	Stereolabs store; Indian resellers e.g. ThinkRobotics
Jetson AGX Orin	NVIDIA authorised distributors; Robu.in, ThinkRobotics, Amazon India
Hall sensors, ICs	Mouser India, DigiKey India, element14 India, Robu.in
Wheel speed sensors	Bosch Motorsport (import); ABS sensors from Indian auto spares market
Encoders	Autonics India, Kübler; hobby encoders from Robu.in
Motors / actuators	maxon India; Indian industrial servo suppliers
Pneumatics (EBS)	Festo India, SMC India, Janatics
Connectors, cables	Mouser, DigiKey, element14, TE / Amphenol distributors

Some practical tips: (1) Plan early, because imported items can take several weeks and must clear customs, and import duty and GST add a large amount to the price. (2) Ask for student, education or sponsorship pricing, since many companies support Formula Student teams in exchange for logos and reports. (3) Check if the item needs a WPC approval (for radio parts such as the RES) before importing. (4) Buy from authorised distributors so that warranty and support are valid. (5) Keep spares of cheap parts like Hall sensors, connectors and fuses.

XVIII. CONCLUSION

A driverless Formula Student car depends completely on its sensors. The Pandar40P LiDAR gives accurate distances to the cones, the ZED 2i camera tells their colour, the VN-200 and IMU give the car's motion, the wheel speed, Hall effect and optical sensors measure what the car itself is doing, and the Jetson AGX Orin runs all the software that turns this data into steering and braking commands through the by-wire systems. Every sensor has its own errors, but most of them can be solved with good calibration, rigid mounting, time synchronisation and sensor fusion. Understanding these sensors is the first step towards building a reliable autonomous system for the team.

ACKNOWLEDGMENTS

I would like to express my sincere gratitude to Formula Manipal for providing the opportunity to complete this task on the sensors used in a driverless car. Working on this report significantly improved my understanding of how perception, state estimation and actuation hardware come together in a real autonomous vehicle. I also thank the Manipal Institute of Technology (MIT), Manipal for providing the academic platform that facilitated this learning exercise.

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